

EXECUTIVE SUMMARY OF THE CONFIDENTIAL PRESENTATION **THE NEW HADRONIC POWER PLANTS**

by [REDACTED]

Coal Electric Power Plants are environmentally unacceptable because they release excessively toxic fumes and green house gases, while causing an alarming depletion of breathable oxygen from our atmosphere. Following millions of dollars of investments over two decades, the new **Hadronic Power Plants** have been developed to resolve these environmental problems via a new patented process of coal combustion via electric arcs submerged within oxygen-rich liquid wastes in **PlasmaArcFlow-TM Recyclers** or **Hadronic Reactors-TM**. The new plants and related reactors are called "hadronic" due to the need of using "hadronic mechanics" for credible calculations on submerged electric arcs due to excessive deviations suffered by conventional theories outlined in the above presentation.

The **energy produced by conventional coal combustion** is essentially that released by the formation of CO (255 Kcal/mole). By comparison, the **energy produced by combustion of coal via electric arcs** is greater. In fact, said arcs decompose liquid molecules by forming a plasma of mostly ionized H, C, O and other atoms. The plasma cools down in the surrounding liquid by producing a large amount of heat acquired by the liquid itself plus a combustible gas, called **MagneGas-TM (MG)**, that bubbles to the surface for collection, and carbonaceous precipitates used for the production of electrodes. In the latter case we have 255 Kcal/mole from CO as in the usual coal combustion, and in addition we have 105 Kcal/mole released in the formation of H₂, 57 Kcal/mole released in the recombination of H and OO into H₂O, 87 Kcal/mole in the combustion of CO into CO₂, and other energy releases identified below.

Electric arcs within stationary liquids are very inefficient because most of the produced H and O gases recombine into H₂O, while CO is burned into CO₂ resulting in an unacceptable content of green house gas before combustion. The new PlasmaArcFlow process has resolved the above shortcomings via the continuous removal of the plasma from the electrodes immediately following its creation.

The **Scientific Efficiency (SE)** of PlasmaArcFlow Recyclers is given by $SE = [Energy\ out]/[Energy\ in] = [HT\text{-}Energy + MG\text{-}Energy]/[Electr.\ Energy + Coal\ Energy] < 1$ and it is smaller than one because of the conservation of energy. However, coal is very inexpensive and its cost is more than covered by the income from the recycling of liquid waste. As a consequence, coal is ignored in the energy efficiency. This yields the **Commercial Efficiency CE** $= [HT\text{-}Energy + MG\text{-}Energy]/[Electr.\ Energy] \gg 1$ that is usually much bigger than one because liquids become superconducting under a closed arc with fractional Ohms, thus requiring minimal energy for their maintenance, while the energy in coal is a large multiple of the electric energy needed by the arc under fractional Ohms. Note that the Commercial Efficiency increases with the increase of pressure, power, temperature and other factors. These main principles have permitted the development of three types Hadronic Power Plants as summarized below.

Hadronic Power Plants of the First type are those in which all of the heat produced by the arc and most of MagneGas produced are used for the self-generation of the DC electricity needed by the arc, while the residual MagneGas is used for the production of green AC electricity for release in the grid. The **Hadronic Power Plants of the Second Type** are those in which the MagneGas produced is catalytically liquefied via a Fisher-Tropsch tower, with release of large additional heat that, when combined with the heat produced by the arc is sufficient for the self-generation of the DC electricity, in which case all of the liquid fuel is used for the production of green electricity.

Note that, following extensive verification lasted for years, the **self-sustained production of green electricity by PlasmaArcFlow Recycles** has been admitted by the US Patent Office in patent number 6,663,752 (see also the additional US patents 6,673,322; ; 6,540,966; and 6,183,604 currently pending in Europe).

The **Hadronic Power Plants of the Third Type** are based on new nuclear syntheses permitted by submerged electric arcs via the physical laws of the new **Controlled Hadronic Fusion** that operates at temperatures intermediate between the **failed cold and hot fusions**. In fact, nitrogen N(7,14) is routinely measured in macroscopic amounts in MagneGas produced from liquids without appreciable nitrogen occlusion, thus suggesting its synthesis from Carbon C(6,12) and two H(1,1), with the secondary release of a positron and of the hypothetical neutrino. Similarly, the synthesis of water in

macroscopic amounts is routinely obtained when processing oil-base liquids C_nH_{2n} that have no oxygen at all, thus suggesting the synthesis of $O(8,16)$ from $N(7,14)$ and two $H(1,1)$.

Hadronic Power Plants of the Third Type are based on the proper enhancement of these nuclear syntheses, in which case the heat produced by the Hadronic Reactors is a source of energy dramatically bigger than that needed for the self-generation of electricity. Green AC electricity is then produced via turbines operated by steam produced from excess heat and the combustion of all produced MagneGas.

Nuclear Reactors are environmentally unacceptable because they produce harmful radiations, while releasing highly radioactive waste. **Hadronic Reactors** are environmentally acceptable because they have been independently certified to produce no radiations at all, while releasing no harmful waste.

Three Hadronic Reactors are currently available for the verification of all the above statements upon suitable investments: a 50 KWh reactor in Verona operating at atmospheric pressure, a 100 KWh reactor in Florida operating at 300 psi (10 bars) and 150° C, and a research reactor available in Kazakhstan.

Preliminary measurements for the **Verona Reactor** in need of systematic verification indicate that the use of 100 Wh = 314 BTU-h of AC electricity corresponding to 90 Wh = 280 BTU-h of DC electricity at the arc produces about 42 L-h of MagneGas with 1050 BTU-h plus about 300 BTU-h of heat for the total of 1350 BTU-h BTU-h. Assuming only a 38% efficiency for a conventional DC generator, the data imply the use of 736 BTU-h for the self-generation of DC electricity and 614 BTU-h of excess heat. By using again a conventional (this time AC) generator with 38% efficiency, we have the production of 38% 614 BTU-h = 233 BTU-h = 74 Wh of green electricity for release in the grid. Since the Verona Reactor has 50 Kwh, the reactor is consequently predicted to be able to generate about 2.5 KWh of AC electricity for release in the grid after AC-DC conversion and other losses out of a nominal production of 3.7 KWh. Measurements for the verifications of these data are strongly recommended for funding.

The above commercial efficiency of the Verona Reactor is multiplied when the same reactor is completed with a Fisher-Tropsch Tower for the liquefaction of MagneGas. In this case the DC electricity is entirely produced by the heat of the reactor and of the tower, leaving the liquid fuel available for the production of electricity (Hadronic Power Plants of the Second Type). The completion of the Verona reactor of this type is strongly recommended for funding.

The Florida Reactor has been built for the construction of a fully operational **Prototype 100 KWh Hadronic Power Plant** via the addition of: a heat exchanger for the production of steam; a DC generator operated by a turbine fueled by steam and MagneGas; and a conventional electric generator built for natural gas and operated with MagneGas. The construction of this prototype plant is strongly recommended for funding since available measurements guarantees the self-generation of AC electricity.

The **Florida Reactor** has been also built for the specific purpose of finalizing the physical laws of the Controlled Hadronic Fusion via systematic measurements of the produced heat with the increase of pressure, power, temperature and other factors. The systematic conduction of these measurements is strongly recommended for funding due to their moderate cost compared to unlimited returns.

In summary, independently verified measurements available in www.magneGas.com have established the capability by hadronic reactors of the simplest possible type, as that in Verona, of self-generating the DC electricity of the arc plus producing excess energy suitable for the production of green AC electricity, according to main feature officially admitted in formal claims by the U. S. Patent Office. These results and their independent verifications provide more than sufficient grounds for investments in the progressive development of the technology because all subsequent versions, such as those with the liquefaction of MagneGas and the nuclear syntheses will multiply the efficiency of the simplest reactors.

As typically the case for all Executive Summaries, readers are discouraged from venturing judgments based on the sole scanning of this extremely condensed summary and are requested to study the complete presentation listed above as well as its vast documentation.

Finally, readers are warned that [REDACTED] shall interrupt any and all cooperation in the event the content of this new technology is presented to academia, due to irreconcilable conflicts between organized and vested interest in old pet theories of the past century, and the need of broader covering theories for any serious resolution of the alarming environmental problems.